

Patient Monitoring System (PMS)

Part 1: Application case description

The project involves the development of a *patient monitoring service* (in short: PMS) for cardiovascular diseases. This document describes the system and its application domain and outlines the existing technological and business constraints. Section 1 provides some background about the domain of e-health and patient monitoring services. Section 2 describes the problem domain in further detail. Then, Section 3 lists the key stakeholders involved in the development of this system, and Section 4 describes a number of additional constraints the system should comply with. Finally, Appendix A lists the functional requirements of PMS as use cases, while Appendix B provides the API specification for the hospital integration.

1 Context

The COVID-19 pandemic has created pressures on national health services that are previously unseen. One of the consequences is that incentives are increased for health care providers to seek for more efficient ways to treat and manage patients in an *extra-muros* context: outside of the hospital premises, while keeping beds maximally reserved for more acute treatments. The recent technological development of telemedicine services, paired with the ubiquity of sensing devices and the Internet-of-Things serves as an additional accelerator to this trend [1].

Cardiovascular diseases. Heart diseases or cardiovascular diseases (CVDs) are a class of diseases that involve the heart or blood vessels (arteries and veins). While the term technically refers to any disease that affects the cardiovascular system, it is usually used to refer to those related to arteriosclerosis (arterial disease). This is a condition in which an artery wall thickens as the result of a build-up of fatty materials such as cholesterol. Arterial disease can lead to heart attacks and strokes. Other pathologies are hypertension, congestive heart failure, etc.

Population-based studies in the youth show that the precursors of heart diseases start in adolescence [2]. CVDs are slow, in the sense that they develop over decades, but as they become more acute in older patients, CVD is often associated to a patients' age. Although some symptoms can be treated (for example, obstructed heart vessels can be bypassed), prevention is of utter importance (for example, change in diet). Smoking, hypertension, and other conditions can increase the risk by several times.

As the western world faces population aging, the impact of CVDs increases continually. Most countries face high and increasing rates of cardiovascular disease. In 2015, cardiovascular diseases (CVD) cost the EU economy over 210

billion, and of this total cost, €111bn (53%) was spent on health care, €45bn (21%) in informal care costs, €32bn (15%) due to early mortality; and €23bn (11%) due to absence from work or early retirement [3]. The priority in CVD treatment is on (i) primary prevention and timely diagnosis of the disease, and to (ii) management of the disease, i.e. prediction and prevention of malignant events (e.g., heart failure) after diagnosis.

E-health and telemedicine. *Electronic health* has large potential for increasing the quality of care delivered to patients, while decreasing overall health care costs. It can therefore mainly contribute to the management of CVDs, less to the primary prevention and timely diagnosis. The term “e-health” is defined broadly, grouping all types of health care practice supported by electronic processes and communication. It ranges from medical knowledge management (medical journals, etc.) to software solutions for appointment scheduling and patient data management [4].

Another instance of e-health is the centralization of patient data in an *electronic health record* (EHR) which enables different health care professionals to communicate patient data in a fast, efficient and reliable way. *Telemedicine* is another instance of e-health, which enables the remote treatment and monitoring of patients, for example at home. As opposed to treatment and monitoring at the hospital, telemedicine has the potential to drastically cut costs in hospitalization.

A patient monitoring service or system focuses not so much on active treatment, but on monitoring of the patient and the disease. It offers extensive and up-to-date information to the physician without being overly obtrusive in the patients’ day-to-day life. By continually monitoring the patient, the physician can predict and therefore prevent malignant events, for example by changing the diagnosis or adapting the treatment.

Positioning

In this assignment, we assume the point of view of a startup software company that enters the market with a novel patient monitoring service¹ (PMS) for the management of cardiovascular diseases (CVD). At this stage of the development, initial collaboration with a local hospital for pilot studies is already arranged. Although this service is being built with one specific hospital in mind, it would evidently be beneficial for our company if we can extend our offering to other hospitals as well.

The goal is to offer the PMS *as a service*: instead of developing, selling or licensing the software as-is, our company will provide and maintain a patient monitoring infrastructure which will interact with the hospital services. The system itself will be located and hosted outside of the hospital infrastructure, but it will be well-integrated in the existing hospital infrastructure or Hospital Information System (HIS), for example to access patient records, offer decision support functionalities to the cardiologist, for invoicing, etc. It will also synchronize data with Electronic Health Record (EHR) services.

Key innovations. Whereas existing systems and services are relatively rigid in how they support the different models (e.g., using static rule-based

¹Although this assignment presents in fact a simplification of reality, the requirements and description of the problem domain are based on a number of actual systems and research prototypes in the e-health space.

decision models), our service will flexibly apply innovative machine learning (ML) algorithms and classifiers [5, 6] (which are shown to outperform traditional approaches [7]) to perform risk estimation on a frequent basis and monitor the evolution of CVD in specific patients.

2 Description of the problem domain

This section discusses the overall goals and requirements of the Patient Monitoring Service (PMS).

2.1 Overall System Goals

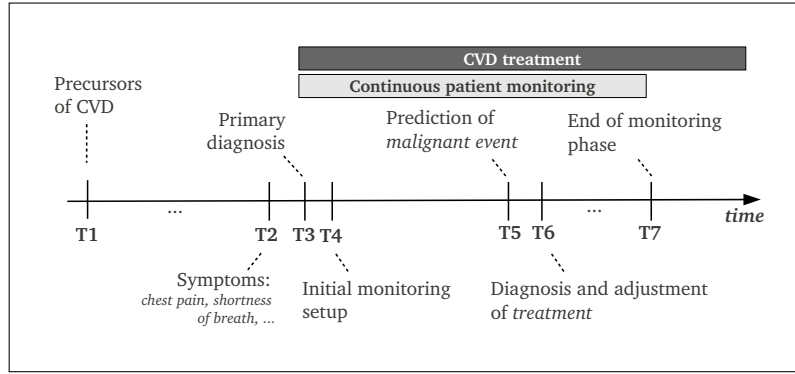


Figure 1: High-level timeline overview of the treatment of a CVD patient.

Figure 1 schematically outlines the typical evolution of CVD. The enrollment of a patient in the CVD monitoring scheme happens after primary diagnosis (T3 in Figure 1), and it ends when it is considered no longer necessary by the physician in the context of overall follow-up and CVD treatment. The main goal of the service is to offer a sound and patient-centric solution for continuous and remote monitoring, in support of timely decision making and accurate prediction of malignant events, without in turn being overly invasive in the patient’s day-to-day life.

- **Continuous and remote monitoring.** Patient monitoring is a very powerful tool in treatment and management of long-term diseases such as CVD. Automated continuous patient monitoring can give physicians a detailed and up-to-date view on the evolution of the disease. Current technologies allow patients to be monitored 24/7 without the need for the patient to be confined in a hospital bed.
- **Timely decision making.** Continuous patient monitoring also aids health care professionals in their decision making, offering them an extensive and up-to-date view on the patient’s status. This leads to improved diagnoses, while reducing the necessity of frequent in-person consultations.
- **Prediction of malignant events.** Furthermore, continuous monitoring of the patient’s health status can predict malignant events such as heart attacks. By

automated prediction of such events, health care professionals can be notified in a timely fashion, and the occurrence of malignant events such as heart attacks can be prevented.

- **Invasiveness in the patient’s day-to-day life.** To ensure acceptability of such a monitoring service, a patient should not be bound to one place for the monitoring to happen.

The remainder of this chapter zooms in on these four aspects of the patient monitoring service.

2.2 Continuous monitoring

To monitor CVD patients, many parameters are of relevance, such as the heart rate, weight, blood pressure, body temperature, etc. In the patient monitoring service, these measurements are provided by electronic sensors, but also directly by the user.

Wearable unit. The most important enabler for monitoring CVD patients is the *wearable unit*. This device bundles a number of sensors for CVD and is worn by the patient close to the heart, for example as a chest band. The wearable unit has a battery lifetime of at least 8 hours and is minimal in size in order to not hinder the patient. The wearable unit is a medical device capable of continually measuring the following medical parameters:

- the **heart rate** in beats per minute (bpm). Normal levels for a healthy person are between 60 bpm and 100 bpm. For correct interpretation, the heart rate values should be correlated with the activity level of the patient.
- a continuous **electrocardiogram** (ECG) which plots the electrical activity of the heart over time (see Figure 2). The fundamental parameters to evaluate are the **maximum ventricular rate** (maximal heart rate) in bpm, the presence of **ventricular arrhythmias** (abnormalities of the heart rate and rhythm) and the presence of ischemia signs (restriction in blood supply).
- the **respiratory rate** (based on sound). Normal levels for a healthy person are within 12 and 20 breaths per minute. This value also needs to be correlated to the activity level of the patient.
- the **blood oxygen level** in percentage. Normal levels for a healthy person are within 90% and 100%.
- the **blood pressure** (the pressure exerted by circulating blood upon the walls of blood vessels) in millimeters of mercury (mmHg). Normal levels for a healthy person are within 85 mmHg and 140 mmHg systolic.
- the **activity level** using the accelerometer. The activity level is expressed as a single number.
- the **body temperature** in degrees Celsius. Normal levels for a healthy person are within 35 degrees and 37 degrees.

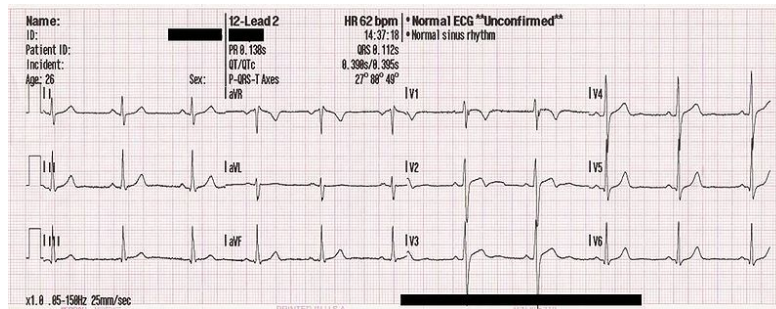


Figure 2: An ECG of a 26-year-old male.

Additional inputs. Next to the wearable unit, support for other measuring devices, sensors or services could be integrated in support of the patient monitoring service. For example, the patient’s weight can be obtained (in kg) with a smart scale. A GPS chip (e.g., in a smartphone) delivers info about the geographic location of the patient. In future implementations of the PMS, the service should also provide support for daily or weekly questionnaires to be filled in by the patient, which allows collecting different types of relevant information (e.g., on the patient’s state of mind, activities, etc.). Many sensor-based lifestyle tracking applications already on the market collect relevant data which may be taken into account during the follow-up and treatment of CVD-related pathologies in a patient.

Patient gateway. Sensors will not communicate directly with the PMS back-end service, but the collected data will go through an intermediate device: the *patient gateway*. For practical reasons, the role of patient gateway will be filled in by the patient’s smartphone and the gateway logic will be supported by a smartphone application specifically developed for the PMS.

Every sensor communicates directly with the gateway, which makes abstraction of specific communication technologies, protocols and channels, but is (most of the time) capable of connecting to the back-end services.

The data collected from the sensors and other inputs are aggregated by the gateway and transmitted to the patient monitoring service. The gateway never transmits raw data to the system, only processed values in packages. The exact contents of a data package and the transmission rate can be configured on the gateway. By default, the configuration is determined by the patient’s risk level (see Section 2.4). For example, in case of low risk, only a single average blood pressure value for the monitoring period is required during the planned transmission. In case of high risk, it can be useful to acquire the maximum, minimum and average daily values. Table 1 gives an overview of the default configuration for high risk patients, which has the highest transmission rates. These default rates can be explicitly overridden by a physician.

Next to periodically transmitting the measurements, the gateway also supports transmitting data on demand. The physician has the ability to request the measurements instantly, in which case the system immediately pushes the request to the gateway. For example, the physician can pull the latest data in full detail to perform a remote consultation.

Parameter	Transmission rate	Derived measure
Heart rate	48/day	Avg, max, min, variance
ECG	24/day	Full ECG of 30 sec.
Respiratory rate	48/day	Avg, max, min
Oxygen level	24/day	Avg, max, min
Blood pressure	48/day	Avg, max, min
Activity level	4/day	Single value
Temperature	6/day	Avg

Table 1: Default configuration for high risk patients.

Apart from synchronizing obtained data from the wearable unit to the PMS, the gateway application will also provide a user interface to patients, allowing them to manage their devices (e.g., consult battery percentage), consult the obtained sensor data and risk values, follow-up on overall treatment, contact their GP or request support. It could even support a social network that allows CVD patients to share experiences.

Impact on day-to-day life. A consequence of the technological advances in low-power wearable sensors and mobile communication networks is that the patient is not bound to one place for the monitoring to happen. Monitoring is transparent and does not prohibit the patient from performing regular day-to-day activities. Obviously, in terms of invasiveness, specific attention is to be spent not only on physical interference but also on data privacy in general (cf. Section 4).

2.3 Risk estimation and notifications

The system uses the monitoring data sent by the gateway to continuously make estimates of the patient’s current CVD health status and the evolution therein. The underlying goal is to predict upcoming malignant events (such as a heart attacks) and keep the appropriate parties (general practitioner, cardiologist, etc.) up-to-date by sending notifications.

To achieve this, patients are divided into *risk levels* (low, medium, and high risk). The physician determines the initial risk level of a patient depending on the primary diagnosis.

After that, the system monitors the patient and continually estimates whether or not a patient’s risk level should change, using *clinical risk models*. These models combine different technologies such as data mining, machine learning, probability models (e.g., Bayesian networks) and ontologies to model the data and derive a risk estimate combining all given inputs. The inputs for the clinical risk models are: (i) the new data sent by the gateway, (ii) the old data in the system, (iii) the pathologies of the patient (as determined by the physician), and (iv) the current risk level. Optionally, artifacts related to previous classifications are also used (e.g., specifically trained classifiers for one patient).

To clarify how this works, we provide the example of a very simple clinical risk model. This model estimates a patient’s risk using *thresholds* for each medical parameter, determined on a per-patient basis. These thresholds determine whether the monitored values are to be considered normal, worrisome

or dangerous. Default values for each of these thresholds depend on the risk level of the patient, but the physician is able to override each of the threshold values. Table 2 shows the default threshold values for the systolic blood pressure parameter.

Risk level	Normal	Worrisome	Dangerous
Low	105–120	90–105, 120–150	80–90, 150–190
Medium	95–135	90–95, 135–160	80–90, 160–190
High	90–150	85–90, 150–170	80–85, 170–190

Table 2: Default thresholds for systolic blood pressure (mmHg).

Notice the wider ranges for patients with higher risk-levels. Because of their cardiovascular conditions, their normal values will be, for example, more elevated, than the normal values of low-risk patients.

A different example is HeartScore [8] which provides an interactive tool targeted at the general population and not at known high-risk CVD patients. It is not suitable for event detection, but it can be support decision-making and is definitely useful for patients as it shows their overall progress, and so integration of HeartScore-like calculations is definitely useful.

It is generally accepted that no single clinical model is suited for all different CVD pathologies, and thus the system will need to support different clinical models. In addition, a single patient’s CVD monitoring schedule is typically supported by multiple clinical models that employ different types of algorithms or are aimed at assessing different aspects. The overall outcome of a risk estimation for a single patient is then based on the combination of the outcomes of the individual clinical models selected for that patient. Some clinical models will be versatile or generic (and thus be deployed and configured system-wide), while others may be very specific to certain CVD pathologies, or even need to be configured on a per-patient basis. In any case, all risk estimations are executed by PMS.

The threshold-based risk model described above is obviously a highly simplistic example, and as discussed earlier, one of the innovative features of the new patient monitoring service is that it will rely on clinical risk models that incorporate more advanced Machine Learning classifiers. These are provided by a third party, an external research organization which is specialized in CVD prediction and releases trained classifiers. This third party will provide trained models for specific pathologies and different risk levels, which can be used in the continuous monitoring of patients. These models can be updated over time. To be able to run these models, the PMS will use a machine learning framework.

The main goal of the patient monitoring service is to aid and support medical decision making, *not* to make actual medical decisions. Only the physician is granted authority to effectively change the risk level of the patient in the system. If the system determines that a patient’s risk level should or may change (based on a risk estimation), the physician is notified in order to assess the estimation and approve, revise or decline it.

In general, there are three notification levels: green, yellow, and red. These levels indicate the degree of urgency or priority with which the notification

should be dealt with. A red notification is sent when the patient’s status changes significantly and requires immediate and urgent attention. A yellow notification is sent when the patient’s status changes non-critically but remains potentially dangerous. A green notification is sent when the patient’s status reverts to a lower risk level. Of course, the exact content of the notification depends on the precise event.

2.4 Physician decision support

The medical models used for the estimation of patient risk levels are also used to aid physicians in their decision making, for example on a consultation. When a physician checks on the status of a patient, the most important and relevant information is shown in a structured and efficient way. For this, the system has to be able to evaluate the patient’s status. Through the implementation of standard clinical models and by applying medical guidelines, routine clinical consultations are made more consistent and informative. For example, the system knows that the physician was recently notified of a malignant event with the patient. Now, the system shows the current risk estimate of the patient, together with all the information relevant to the event (for example, a graph of the blood pressure, or an ECG). Of course, physicians are still able to request any additional information they want.

2.5 Patient support

Similarly to physicians, patients can also make use of the medical models. When patients want to check their own status, they also expect to see the most important and relevant information in a structured and efficient way. Of course, there is an important difference in the way this data is presented to these actors: in case of the patient, this data must not be too technical or clinical. The system offers patients a customized view on their data, for example a simple graph indicating the percentage improvement since last week.

2.6 Medical emergencies

As an extension to the green, yellow and red notifications, the system can also send *emergency notifications*. An emergency notification is sent to the Hospital Emergency services (cf. Section 4) in case of a life-threatening event such as a heart attack. Emergency notifications are sent in case of *ongoing* emergencies, while other notifications are sent when malignant events are *predicted* or anticipated.

Regular notifications are sent by the system after analyzing new data using medical risk models. For emergencies, this system is too slow since the fastest transmission rate is 48 times per day. Therefore, the gateway is extended with a general-purpose, lightweight, medical emergency model, which is also threshold-based (as the one presented in Section 2.3). Similar to the other risk models, default threshold values depend on the patient’s risk level but can be explicitly overwritten by a physician.

The gateway uses this model to interpret the measurements itself and alerts the system of a possible emergency, along with the necessary data. The system re-checks the data for an emergency using more extensive models and sends out

an emergency notification if needed. Note however, that we are not building a life-saving device: handling emergencies is not a primary goal of the system, and the timely delivery of an emergency notification should not be guaranteed in every case.

3 Main stakeholders

Patients. The system applies to patients who have already been diagnosed with CVD and now need to be monitored for treating or controlling their disease (which does not hold for every CVD patient). They want the system to help them in the treatment and management of the disease and would like this to happen in the most comfortable way.

Physicians. The system will be used by different types of physicians. In practice, cardiovascular disease is treated by specialists such as cardiologists, thoracic surgeons, vascular surgeons, neurologists, and radiologists, depending on the organ system that is being treated. Another important stakeholder in this context is also the patient's general practitioner (GP) who keeps a holistic overview of overall patient health.

The system should support these physicians in their job, primarily by aiding their decision making. Physicians want the information they need in a structured and efficient way, for example by being notified of important updates of a patient's status or a clean graph of the patient's evolution. Also note that we are building a system for use by a hospital, but not every physician needs to be in that same location. A GP typically has their own independent practice, and often specialists also hold practice outside of hospital premises.

Nurses. Nurses are responsible for the more practical part of the patient's treatment. In general, nurses at the hospital do not take part directly in the system. However, a specialized team is responsible for the setup and configuration of the sensors after a patient has been diagnosed and recommended to use the system.

Emergency call center. The emergency call center in the hospital receives all emergency calls and responds appropriately, for example by dispatching an ambulance to the scene. They also receive emergency notifications from the monitoring service. With an emergency, they want to receive as much information as possible to correctly assess the situation and the urgency to determine the most appropriate response. Therefore, they also want to access the detailed monitoring information. Moreover, an emergency notification also needs to be entered in the system so other appropriate parties (the specialist, the GP etc.) can be notified to be kept up-to-date.

Telemedicine operators. The telemedicine operators deal with issues and provide technical and setup support, ranging from patients who notice that their wearable unit is malfunctioning, to a hospital administrator who notices that data updates are not working correctly.

The telemedicine operators also follow up the overall management of the system. They integrate risk models and deal with updates and upgrades of the

functionality, provisioning system resources, etc. They are notified by the system when a patient’s sensor has not sent in data for a long time or the data seems incorrect (e.g., impossible outliers). They handle notifications where further human assessment is needed to determine further actions.

The hospital. As mentioned in Section 1, our company’s first client is a specific hospital. More specifically, we –as a telemedicine company– have been contracted for a pilot project, involving only a limited number of CVD patients. If successful, the hospital will deploy and apply this monitoring service to a larger patient base in the future.

The hospital’s main stake is that the patient monitoring service will aid them in raising the quality of CVD treatment, thereby increasing the efficiency and effectiveness of CVD treatment. Furthermore, by introducing CVD patient monitoring from home, CVD patients are kept out of the hospital and this frees up space and resources allocated previously for on premise patient monitoring.

Electronic health record service The Electronic Health Record (EHR) service collects patients’ health records from diverse health organizations and provides different physicians with an efficient means to consult patient data while also providing patients with access to their own records. This service can be performed by an individual hospital itself or by a specialized third party, for example one service used by all hospitals and physicians in a geographic region. The data collected by the patient monitoring service will eventually be integrated into the patient’s EHR and kept up to date via the HIS.

Legal departments. Both the legal department of our company as well as that of the hospital need the system to comply to health care regulations. More information about the precise health care regulations can be found in Section 4.3. It is important the system is not only compliant to these regulations by design but is provably compliant throughout its operation.

Hospital financial department. The hospital financial department is responsible for invoicing their customers, the patients. It is imperative that the systems provide clear usage and billing reports so that this is done correctly.

Telecom operators. Telecom operators provide means for the sensors and the gateways to communicate with the system. A service level agreement (SLA) between our company and the telecom operators determines the capabilities of these communication channels.

ML model provider. This stakeholder represents an external clinical research institute that trains and prepares specific Machine Learning models tailored to specific pathologies under the broader CVD umbrella. The institute’s clients can download the trained models and classifiers and allowing them to use these models on their own medical data, such as the collected sensor measurements in case of PMS. Furthermore, this stakeholder also prepares regular updates and improvements to the trained models and classifiers they provide. In order to simplify their client’s use and management of trained models this stakeholder provides a ML framework that can be integrated into existing systems. This

framework provides support to execute trained models as well as communicate with the stakeholder’s system to, for example, update models or obtain new ones.

4 Key building blocks and constraints

Next to the general problem domain description, there are a number of specific constraints highly relevant to the patient monitoring service.

4.1 Hospital Information System (HIS)

As mentioned in Section 1, the patient monitoring service will be deployed and maintained as a standalone platform by our company. Nonetheless, it should integrate tightly with the existing Hospital Information System (HIS) currently in use at the hospital. A relevant resource in this specific respect is the HL7 FHIR (Fast Healthcare Interoperability Resources) standard² which is widely adopted in the e-health domain. The desired integration covers a number of dimensions:

Patient data. Evidently, the hospital retains full control of patient records and patient data, and the patient monitoring service (PMS) will not change this³. This implies that the relevant measurements gathered by the PMS must be integrated in the patient record storage of the hospital itself. However, keep in mind that some of the measured data (cf. Section 2.2) is very fine-grained and updated frequently, making it infeasible to forward and store all raw data in hospital’s patient records. The synchronization of patient data between both parties is an important functional aspect of the PMS.

Physician workstation. As part of the existing HIS, the physician (e.g., a cardiologist) already has a means to connect to the hospital, either through their personal workstation or remotely (if they have their own practice), to consult patient records, extend notes of patient consultations, etc. As the patient monitoring service offers the physician with a number of new functionalities related to decision support, remote configuration of the sensors, daily patient follow-up and receiving notifications, these must be integrated in the physician’s workstation.

Patient registration. When a patient is first introduced to the monitoring service, they receive a wearable unit containing a set of sensors. First, these must be paired or linked to the specific patient. Secondly, they must be initialized and calibrated according to the patient’s context, and this setup must be tested before sending the patient home. This process of patient registration happens at the hospital (e.g., after a check-up), and is conducted by a trained nurse.

²The standard specifications can be found at <https://hl7.org/fhir/>.

³From a regulatory perspective [9], the hospital is the *data controller* that determines the means of the data collection activity and the PMS is a *data processor* that facilitates and contributes to the data collection activity organized by the hospital

Emergency situation. When the automated decision support identifies an emergency situation (cf. Section 2.3), it must be able to trigger an emergency procedure by contacting the emergency services already offered by the hospital (e.g., 112) and possibly also the other stakeholders involved in the patient’s treatment, such as his physician and GP.

Accounting. Finally, for purposes of accounting, the hospital must be kept aware on a day-to-day basis of the costs incurred during operation of the patient monitoring service. Typical costs include the maintenance of the platform (e.g., repairing sensors), computational cost (e.g., to perform risk assessment), storage costs, bandwidth costs, and operation costs of the call-center.

4.2 Financial constraints

Evidently, setting up and maintaining a pilot project such as the one described in this assignment is a costly operation. For this project, our company has negotiated a million-euro contract with one specific hospital. Since the pilot project involves monitoring of actual CVD patients, a Service-Level Agreement (SLA) has been enclosed as part of this contract. This agreement stipulates certain quality requirements (in terms of performance and availability). For example, there should be an upper limit in the time it takes for an emergency notification to reach its destination.

Also, our company has to negotiate a number of service-level agreements with several third parties: the telecom operator to ensure the desired degree of patient connectivity, the hardware manufacturer for acquisition, maintenance, upgrade and repair, etc.

4.3 Legal constraints

Because the data handled in the system is medical data, the system has to comply to several laws. In the EU, medical information is regarded as personal information and falls under the General Data Protection Regulation [9] (GDPR). Without going into too much detail, the main guiding principles are:

- Health information cannot be collected and used without explicit patient consent, and in this case only for purposes of continuous CVD treatment, unless processing is necessary to protect the vital interests of the patient (e.g., emergencies), payment or administration of the service.
- Disclosure of identity information should be minimized: if the data can be anonymized or de-identified, it should be (e.g., when opening up the data for research purposes).
- Patient should be able to locate and view all of their health information. They should also be able to obtain, delete, and modify all of their personal that is processed by PMS (right to access, right to rectification, right to removal, etc.).
- Health data should be stored for a number of years, depending on the country (30 years in Belgium, 6 years in the US).
- Accountability: anytime the patient’s data is disclosed to a third party, this event should be monitored and logged.

The system should comply with at least all rules stated above.

4.4 Security

Considering the nature of the patient monitoring service (patient health information is considered sensitive information) and the legal implications listed above, it is evident that security is key concern in the system. In this assignment we limit the scope to *user authentication* and *user action logging*.

User authentication. Authentication is the process of ensuring the identity of the different actors. It should not be possible for a user to act as another user (spoofing) and users should not be able to access the system unauthorized. It is clear that the patient monitoring service has a large variety of types of users, and it is realistic that the concrete realization of authentication will differ highly between these users: for example, physicians may use a badge to authenticate themselves, whereas patients may be authenticated implicitly, because they are using a wearable unit that is linked to his identity.

User action logging. Logging is the process of storing information about every action performed by the actors and the system itself. It enables the checking of system compliance to business or security policies, or in case of security-related incidents, trace the incident to a user. A log contains information such as the identity of the actor, the action, the object on which the action is performed, a timestamp etc. Note that user action logging depends on correct user authentication.

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